

Transfusion Ratios and Deficits in Injured Children With Life-Threatening Bleeding*

OBJECTIVES: To assess the impact of plasma and platelet ratios and deficits in injured children with life-threatening bleeding.

DESIGN: Secondary analysis of the MAssive Transfusion epidemiology and outcomes In Children study dataset, a prospective observational study of children with life-threatening bleeding events.

SETTING: Twenty-four children's hospitals in the United States, Canada, and Italy.

PATIENTS: Injured children 0–17 years old who received greater than 40 mL/kg total blood products over 6 hours or were transfused under activation of massive transfusion protocol.

INTERVENTION/EXPOSURE: Weight-adjusted blood product volumes received during the bleeding event were recorded. Plasma:RBC ratio (plasma/RBC weight-adjusted volume in mL/kg) and platelet:RBC ratio (platelet/RBC weight-adjusted volume in mL/kg) were analyzed. Plasma deficit was calculated as RBC mL/kg – plasma mL/kg; platelet deficit was calculated as RBC mL/kg – platelet mL/kg.

MEASUREMENTS AND MAIN RESULTS: Of 191 patients analyzed, median (interquartile range) age was 10 years (5–15 yr), 61% were male, 61% blunt mechanism, and median (interquartile range) Injury Severity Score was 29 (24–38). After adjusting for Pediatric Risk of Mortality score, cardiac arrest, use of vasoactive medications, and blunt mechanism, a high plasma:RBC ratio ($> 1:2$) was associated with improved 6-hour survival compared with a low plasma:RBC ratio (odds ratio [95% CI] = 0.12 [0.03–0.52]; $p = 0.004$). Platelet:RBC ratio was not associated with survival. After adjusting for age, Pediatric Risk of Mortality score, cardiac arrest, and mechanism of injury, 6-hour and 24-hour mortality were increased in children with greater plasma deficits (10% and 20% increased odds of mortality for every 10 mL/kg plasma deficit at 6 hr [$p = 0.04$] and 24 hr [$p = 0.01$], respectively); 24-hour mortality was increased in children with greater platelet deficits (10% increased odds of 24-hr mortality for every 10 mL/kg platelet deficit [$p = 0.02$]).

CONCLUSIONS: In injured children, balanced resuscitation may improve early survival according to this hypothesis generating study. Multicenter clinical trials are needed to assess whether clinicians should target ratios and deficits as optimal pediatric hemostatic resuscitation practice.

KEY WORDS: blood transfusion; child; deficit; hemorrhage; ratio

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Bleeding is an important cause of preventable death in children (1–3). Reported 28-day mortality rates in children with life-threatening bleeding range from 36% to 50% (4, 5), substantially higher than the mortality rates for nonbleeding pediatric cohorts with comparable illness severity (8–16%) (6–8) and the mortality rates reported in injured adults with life-threatening bleeding (21–24%) (9, 10). Hemostatic resuscitation has evolved over recent decades to include early blood product transfusion, avoidance of crystalloid, prevention of

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RESEARCH IN CONTEXT

- Timely transfusion is a key component of hemostatic resuscitation in traumatically bleeding children; optimal pediatric resuscitation guidelines have not been established.
- There are conflicting data regarding the impact of plasma:RBC and platelet:RBC ratios in pediatric trauma resuscitation; these data are often limited by retrospective collection and survival bias.
- This prospective observational study will examine the impact of blood product ratios and deficits on survival in traumatically bleeding children.

coagulopathy, hypothermia, hypocalcemia and acidosis, and surgical or interventional control of bleeding when applicable (11, 12). In adult trauma patients, the ratio of plasma to RBCs transfused (volume plasma/volume RBC ratio) and the ratio of platelets to RBCs transfused (volume platelet/volume RBC) have been the focus of much research. In several adult studies, high (> 1:2) ratio resuscitation has been associated with improved survival and other outcomes (9, 13–17) and is generally considered to be the standard of care for optimal adult resuscitation practice (11).

There are little pediatric-specific data to guide optimal pediatric hemostatic resuscitation. In pediatric trauma patients, the data regarding ratios are not conclusive; although some studies comparing high ratio with low ratio blood product resuscitation show a benefit to high ratio resuscitation (18–21), several other studies show no mortality benefit (22–26). Importantly, ratios do not account for the volumes of products administered. Plasma deficit (RBC volume – plasma volume) and platelet deficit (RBC volume – platelet volume) provide additional insight into the balance of resuscitation and may be more sensitive markers of efficacy than ratios (27). These studies suffer from small sample sizes which limits both generalizability and power to detect a difference in outcomes. The primary objective of this study was to assess the impact of balanced resuscitation according to blood product ratios and deficits on 24-hour survival for injured children with life-threatening bleeding.

MATERIALS AND METHODS

Research Ethics

The Massive Transfusion In Children (MATIC) study was conducted under waiver of informed consent granted by each participating center's institutional review board (IRB). IRB approval numbers by center can be found in **Supplemental Digital Content** (<http://links.lww.com/PCC/B987>).

Study Design and Data

A full description of the MATIC study has been published recently (4). Briefly, this was a prospective, observational study of children with life-threatening bleeding at 24 medical centers in the United States, Canada, and Italy from January 2014 to October 2018. Inclusion criteria were children 0–17 years old who received greater than 40 mL per kg of total blood products over 6 hours with no break in blood product administration greater than 30 minutes or received a blood transfusion under massive transfusion protocol (MTP) activation (28). In order of priority, time of life-threatening hemorrhage (LTH) initiation was defined based on clinical documentation of hemorrhage initiation, time first blood product requested from blood bank, time first blood product dispensed from blood bank, or time first blood product administered to patient. Time of LTH conclusion was defined by clinical documentation of hemorrhage cessation, time MTP was discontinued, or the time the last blood product was administered. Patients with multiple life-threatening bleeding events were excluded from the study. Data extracted included demographics, etiology of bleeding (trauma, operative, and medical), Glasgow Coma Scale, vital signs, measures of end-organ function, fluids, blood products, hemostatic adjuncts, Pediatric Risk of Mortality (PRISM) III score (29), Injury Severity Score (ISS), morbidities, and mortality. Missing weight values were estimated using the 50th percentile based on age and sex from the Centers for Disease Control and Prevention Clinical Growth Charts. Cause of death was determined by the participating sites.

Study Cohort

This was a planned secondary analysis of the MATIC study database. A subgroup of children from the MATIC cohort with traumatic injuries as the etiology

of their bleeding event was selected. Children were excluded if MTP was activated, but no blood products were administered or if no RBCs were transfused which would not allow for ratios to be calculated. Patients were excluded from the plasma:RBC analyses if plasma was not transfused and from the platelet:RBC analyses if platelets were not transfused.

Analysis

Data were summarized as counts, median (interquartile range), or percentage. Transfusion volumes that were given during the life-threatening bleeding event only were recorded and standardized by subject weight (mL/kg). Blood products given prior to when life-threatening bleeding started or after it had ended were not included in ratio or deficit calculations, in order to directly assess the resuscitation approach during the event. Plasma:RBC ratio was calculated as the weight-adjusted volume of plasma divided by weight-adjusted volume of RBCs transfused; platelet:RBC ratio was calculated as weight-adjusted volume of platelets divided by weight-adjusted volume of RBCs transfused. Patients were categorized into those who received either high or low plasma:RBC ratios and platelet:RBC ratios; a high ratio was defined as above 0.5 or 1:2 plasma:RBC as this was the cut point that optimized sensitivity and specificity to predict 6-hour (0.5), 24-hour (0.6), and 28-day (0.5) survival based on the Liu method of cutpoint estimation. Plasma deficit was calculated as RBC mL/kg – plasma mL/kg; platelet deficit was calculated as RBC mL/kg – platelet mL/kg.

The main outcome was 24-hour mortality from the onset of the life-threatening bleeding event (30). Secondary outcomes included 6-hour and 28-day mortality as well as in-hospital complications (acute respiratory distress syndrome [ARDS], sepsis, and acute kidney injury [AKI]). Bivariate analysis was performed to identify variables associated with 6-hour, 24-hour, and 28-day mortality. Variables with no more than 10% of values missing were included in the bivariate analyses. Multivariable logistic regression models were then generated for mortality at these same timepoints considering all variables with p value of less than 0.2 on univariate analysis, which included age, sex, PRISM score, ISS, and mechanism of injury, as well as use or occurrence of the following during the life-threatening bleeding event: mechanical ventilation, cardiac arrest, vasoactive drugs, blood products, hemostatic adjuncts,

and IV fluids. Complete cases were included. A backward stepwise covariate selection technique was used to generate a parsimonious model and avoid overfitting. Covariates with evidence of collinearity with other potential confounders defined as r greater than 0.6 were not included in the model. All hypothesis tests were two sided, and a p value of less than 0.05 was considered significant. Statistical analyses were conducted using Stata Statistical Software, Version 16 (StataCorp, College Station, TX).

RESULTS

Of 191 patients analyzed (**Fig. 1**), median (IQR) age was 10 years (5–15 yr), 61% were male, 61% blunt mechanism, and median (IQR) ISS was 29 (24–38) (**Table 1**). The cumulative mortality rates for the cohort were as follows: 17% ($n = 33$) at 6 hours, 24% ($n = 45$) at 24 hours, and 37% ($n = 69$) at 28 days. In-hospital complication rates were as follows: ARDS 16%, sepsis 8%, and AKI 6% (**Table 2**). The median (IQR) total blood product volume transfused during the life-threatening bleeding event was 50 mL/kg (28–92 mL/kg). The median (IQR) plasma:RBC ratio was 0.5 (0.2–0.8), median (IQR) platelet:RBC ratio was 0.2 (0–0.3), median (IQR) plasma deficit was 12 mL/kg (5–27 mL/kg), and median (IQR) platelet deficit was 23 mL/kg (11–43 mL/kg) (**Table 2**).

Blood Product Ratios

In total, 100 of 191 (52%) were in high ratio plasma:RBC group, and 23 of 191 (12%) were in the high ratio platelet:RBC group. There were no differences in markers of disease severity or demographics between high and low plasma:RBC and platelet:RBC groups. The high ratio plasma:RBC group more often received cryoprecipitate (31% vs 17%; $p = 0.03$), recombinant factor VIIa (11% vs 4%; $p = 0.02$), and tranexamic acid (14% vs 4%; $p = 0.02$), as well greater mean platelet volumes (9 vs 2 mL/kg; $p < 0.001$). The patients in the high ratio platelet:RBC group had a greater percentage of group A and group B and Rh-negative blood type, and more often received recombinant factor VIIa (26% vs 7%; $p = 0.004$), tranexamic acid (26% vs 7%; $p = 0.004$), and on average greater volumes of plasma (22 vs 13 mL/kg; $p = 0.08$).

Associations between 6-hour, 24-hour, and 28-day mortality and various covariates, including plasma:RBC

TABLE 1.
Injured Children With Life-Threatening Bleeding: Characteristics of the Cohort

Variable	N	Value
Demographics		
Age (yr), median (interquartile range)	191	10 (5–15)
Male, <i>n</i> (%)	191	116 (61)
Race, <i>n</i> (%)	191	
White		91 (57)
Black		65 (41)
Other		3 (2)
Hispanic, <i>n</i> (%)	166	16 (10)
Blood group, <i>n</i> (%)	182	
A		63 (35)
B		25 (14)
AB		5 (3)
O		89 (49)
Rhesus factor +, <i>n</i> (%)	181	149 (82)
Injury characteristics, <i>n</i> (%)		
Mechanism	191	
Blunt		116 (61)
Penetrating		63 (33)
Blunt and burn		1 (0.5)
Blunt and penetrating		11 (5.5)
Injury Severity Score, median (interquartile range)	171	29 (24–38)
Prelife-threatening hemorrhage variables		
Pediatric Risk of Mortality III score, median (interquartile range)	178	11 (5–23)
Age-adjusted hypotension, <i>n</i> (%)	187	102 (55)
Age-adjusted tachycardia, <i>n</i> (%)	172	103 (60)
Mechanical ventilation, <i>n</i> (%)	180	97 (54)
Lowest Glasgow Coma Scale score, median (interquartile range)	159	3 (3–13)
Lowest temperature, median (interquartile range)	115	36 (35–36.5)
Lowest hemoglobin, median (interquartile range)	176	9.9 (8.4–12.1)
Lowest platelet count, median (interquartile range)	131	209 (149–281)
Highest international normalized ratio, median (interquartile range)	92	1.5 (1.3–1.9)
Fibrinogen, median (interquartile range)	119	182 (142–237)

and platelet:RBC ratios, are listed in **Supplemental Table 1** (<http://links.lww.com/PCC/B988>). On unadjusted analysis, high plasma:RBC ratio was associated with increased survival at 6 hours post bleed and approached significance at 24 hours post bleed (**Fig. 2**) (Supplemental Table 1, <http://links.lww.com/PCC/B988>). There was no difference in cause of death (hemorrhage vs traumatic brain injury) between high and low

ratio groups on unadjusted analysis at any time point (all $p > 0.22$). Regarding in-hospital complications, only the rate of ARDS was greater among children transfused with a high plasma:RBC ratio compared with a low plasma:RBC ratio (21% vs 10%; $p = 0.04$) on unadjusted analysis (data not shown). The covariates included in the final ratio models are shown in **Supplemental Table 2** (<http://links.lww.com/PCC/B989>). In these adjusted

TABLE 2.
Blood Products and Adjuncts Transfused During the Bleeding Event and Clinical Outcomes

Variables	N	Values
Blood products and adjuncts		
Duration of massive transfusion protocol (min), median (interquartile range)	190	188 (85–345)
Total volume, mL/kg, median (interquartile range)	191	50 (28–92)
RBC volume, mL/kg, median (interquartile range)	191	28 (16–53)
Plasma volume, mL/kg, median (interquartile range)	191	14 (5–28)
Platelet volume, mL/kg, median (interquartile range)	191	4 (0–14)
Cryoprecipitate volume, mL/kg, median (interquartile range)	191	0 (0–0)
Crystalloid volume mL/kg, median (interquartile range)	191	27 (2–59)
Tranexamic acid (yes/no), <i>n</i> (%)	191	19 (10)
Factor VII (yes/no), <i>n</i> (%)	190	15 (8)
Deficits and ratios, median (interquartile range)		
Plasma deficit, mL/kg	191	12 (5–27)
Platelet deficit, mL/kg	191	23 (11–43)
Plasma:RBC ratio	191	0.5 (0.2–0.8)
Platelet:RBC ratio	191	0.2 (0–0.3)
Outcomes, <i>n</i> (%)		
Mortality	189	
Died within 6 hr		33 (17)
Due to hemorrhage		20 (61)
Died within 24 hr		45 (24)
Due to hemorrhage		20 (44)
Died within 28 d		69 (37)
Due to hemorrhage		42 (61)
In-hospital complications	191	30 (16)
Acute respiratory distress syndrome	191	15 (8)
Sepsis	191	12 (6)
Acute kidney injury		

analyses, higher plasma:RBC ratios were independently associated with reduced 6-hour mortality (odds ratio [95% CI] = 0.12 [0.03–0.52]; $p = 0.004$). The plasma:RBC ratio was not associated with mortality at 24 hours and 28 days. Platelet:RBC ratio was not associated with mortality at any time point.

Blood Product Deficits

On univariate analysis, both plasma deficit and platelet deficit were significantly associated with 6-hour, 24-hour, and 28-day mortality (Supplemental Table 1, <http://links.lww.com/PCC/B988>). The covariates included in the

final deficit models are shown in **Supplemental Table 3** (<http://links.lww.com/PCC/B990>). Both 6-hour and 24-hour mortality were independently associated with greater plasma deficits. The odds of mortality increased 10% and 20% for every 10mL/kg plasma deficit at 6 hours ($p = 0.04$) and 24 hours ($p = 0.01$), respectively. Only 24-hour mortality was independently associated with greater platelet deficits. The odds of 24-hour mortality increased 10% for every 10mL/kg platelet deficit ($p = 0.02$). There was no significant association between plasma or platelet deficit and mortality 28 days post bleeding event (Supplemental Table 3, <http://links.lww.com/PCC/B990>).



AT THE BEDSIDE

- Children with traumatic bleeding are at high risk of death.
- In this study, balanced resuscitation was associated with improved 6- and 24-hour survival in injured children.
- Optimal hemostatic resuscitation practice remains uncertain; further pediatric-specific data are needed.

DISCUSSION

In this secondary analysis of an international, multicenter prospective pediatric study, injured children who received high ratio ($> 1:2$) plasma:RBC resuscitation during their bleeding event had reduced 6-hour mortality. Further, plasma deficit and platelet deficit were independently associated with increased 6-hour and 24-hour mortality in the setting of life-threatening bleeding following trauma. These data are presented from the first prospective study in children specifically designed to investigate resuscitation and outcomes after LTH. Children with life-threatening bleeding have a very high mortality rate (5); identifying therapeutic interventions, such as a balanced resuscitation that can be applied rapidly upon presentation, may improve outcomes for children hemorrhaging after injury.

Few pediatric studies report mortality outcomes related to blood product ratios (18–20, 22–26). Although many of these small, retrospective studies found no significant difference in mortality based on plasma and platelet ratios among children undergoing massive transfusion, some did report improved survival for children transfused to a high plasma:RBC ratio (18–21). In the present cohort, 6-hour mortality was reduced in the subgroup receiving high plasma:RBC ratio resuscitation compared with low plasma:RBC ratio resuscitation. Most of the reports referenced above did not delineate the timeline for mortality; thus, it is possible that any early mortality improvements present in these studies were concealed when grouped in with overall mortality. In a recent consensus conference hosted by the National Institute of Health and U.S. Department of Defense, 6-hour or 24-hour mortality was identified

as important and clinically relevant efficacy outcomes for interventions in severely bleeding trauma patients, whereas 28-day mortality serves as a safety outcome (30). In the current cohort, there was no difference in 28-day mortality. Notably, late mortality may be influenced by circumstances unrelated to hemorrhage, namely sepsis, concomitant head injury, and multisystem organ failure, and is therefore not suited to reflect the effectiveness of early hemostatic resuscitation.

Platelet ratios were not associated with survival at any timepoint. There was substantial crossover among the high ratio plasma and high ratio platelet groups, meaning those who received high volumes of plasma were also more likely to have received high volumes of platelets. It is challenging to isolate the impact of one ratio versus the other, particularly in this cohort where the administration, volumes, and ratios of platelet transfusion were low overall. Further study should identify the impact, timing, and barriers to platelet transfusion to inform optimal pediatric resuscitation practice.

In this study, plasma and platelet deficits were both independently associated with improved early survival in bleeding children after injury. In a 2011 study in 438 injured adults who received 1 or more RBC units in the first 24 hours, plasma deficit and plasma ratio were assessed as predictors of mortality (27). Mortality significantly increased with greater plasma deficit status but not with plasma ratio; interestingly, most of this effect was seen within the first 2 hours of resuscitation, supporting contemporary data that the timing of plasma administration influences patient outcomes (14, 31). There is little research regarding deficits as a prognostic indicator or treatment target in pediatric trauma cohorts. Although plasma ratio was associated with 6-hour mortality only, plasma deficit was associated with both 6- and 24-hour mortality. Further, platelet deficit but not platelet ratio was associated with increased mortality at 6 and 24 hours. This suggests that plasma and platelet deficits, which report not only the balance but also the magnitude of the relationships between various blood products, provide additional and perhaps more useful information as compared to standard product ratios in injured children. Plasma and platelet deficits should be considered for inclusion in future hemostatic resuscitation studies; their utility as covariates, treatment targets, and outcomes in children receiving massive transfusion for LTH requires further investigation.

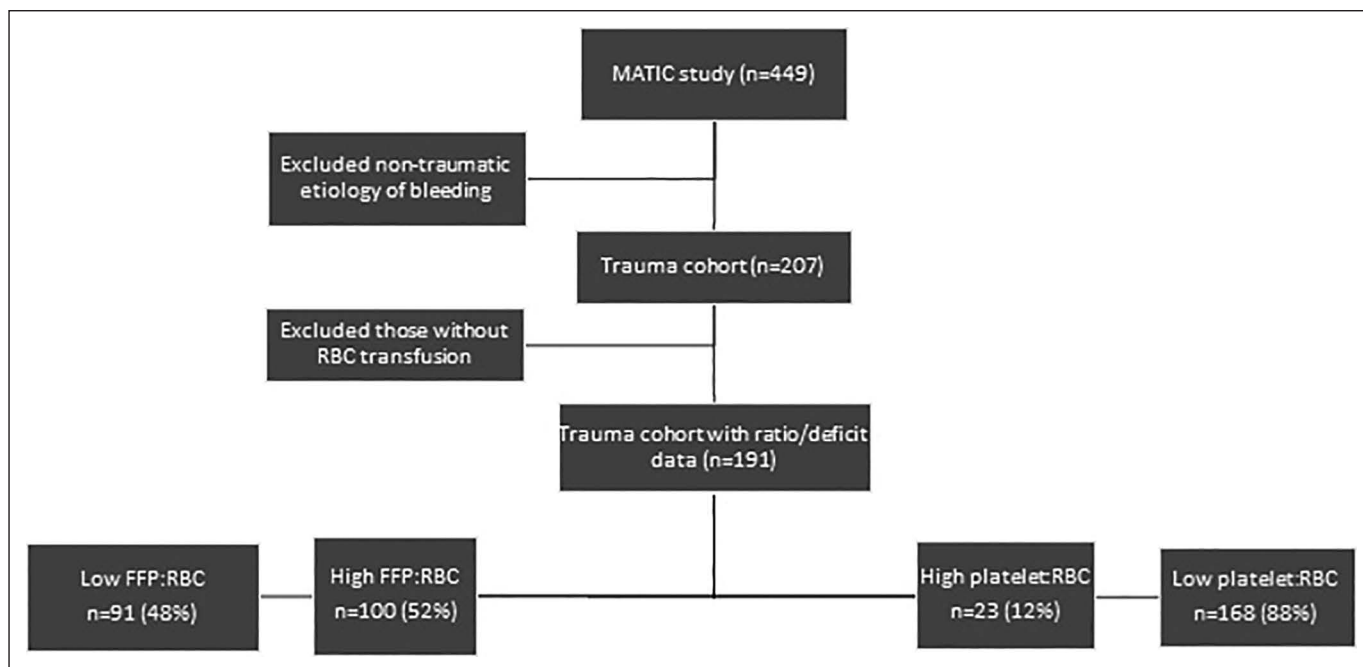


Figure 1. Flow diagram of study participants. FFP = fresh frozen plasma, MATIC = Massive Transfusion In Children.

Children are known to have quantitative and qualitative differences in the hemostatic system as compared to adults at baseline (32, 33). These include decreased plasma concentrations of plasminogen, tissue

plasminogen activator, and α -antiplasmin; increased plasma concentrations of plasminogen activator inhibitor-1; and a decrease in both plasmin generation and overall fibrinolytic activity (32). Further, the lev-

els of protein C, protein S, seven coagulant proteins (II, V, VII, IX, X, XI, XII), and prothrombin are significantly lower in children than in adults (33). These differences may effect a unique pediatric trauma-induced coagulopathy (TIC) after injury (34), a cascade caused shock, tissue injury, and ultimately hemostatic failure that remains poorly understood in pediatric trauma patients (35, 36). It is not known at what age a child's response to injury resembles that of an adult; additional studies across a spectrum of ages are warranted to assess the impact of age on both TIC and response to therapeutic interventions.

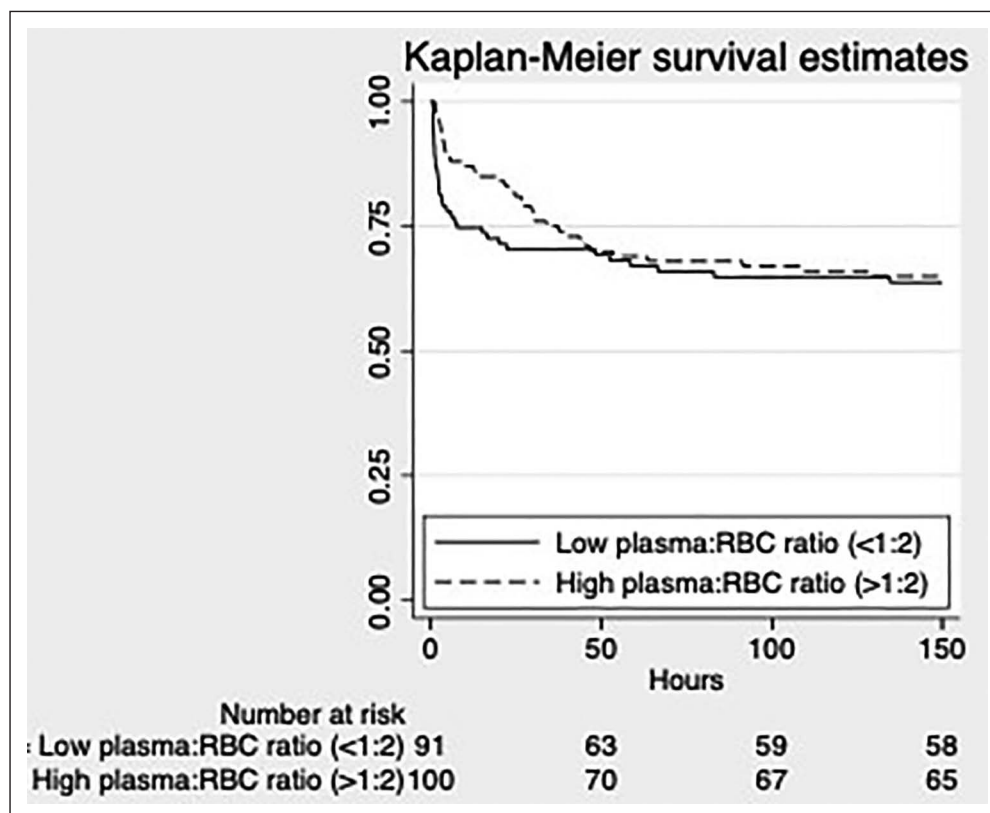


Figure 2. Unadjusted Kaplan-Meier survival curves for high ratio versus low ratio plasma:RBC groups

The independent associations between reduced 6-hour mortality with low plasma deficit and increased plasma:RBC ratio among trauma patients may be related to the effect of early plasma administration on TIC. Some studies suggest that the conversion of protein C to activated protein C via thrombomodulin-thrombin complex formation after endothelial damage drives much of the subsequent coagulopathy by deactivating factors V and VIII (37–40). Considering this in addition to the delayed maturation of the protein C pathway in children, the administration of plasma might be particularly beneficial in this population to ameliorate or halt the effects of TIC in the acute phase of a life-threatening bleed, leading to the observed decrease in mortality at 6 hours. Further study that includes TIC biomarkers in injured children with life-threatening bleeding may offer insight into the mechanism associated with improved survival in this cohort.

Strengths of the study include the prospective multicenter design, the use of weight-adjusted product volumes in mL/kg as is conventional in children, and an analysis that reflects the acute resuscitation solely during the life-threatening bleeding event and is not confounded by other product administered during the hospitalization. Limitations of this study include those inherent to a secondary analysis of existing data, although the prospective observational study was designed to record the variables of interest in this analysis: time and volume of products administered to children during a life-threatening bleeding event. There is variability among centers regarding transfusion protocols, product availability, and volume of transfusion products, although reporting of weight-adjusted volumes does standardize this in part. Other limitations include the size of the cohort and relative infrequency of the outcome of mortality; multicenter studies should be pursued to confirm these findings in larger cohorts. Missing data may predispose to bias; however, missingness was less than 10% on key variables in the model (number analyzed is reported in Table 1 for each variable) and presumed to be at random. Finally, studies of transfusion over time are at risk of survivor bias, meaning that it is possible that the survival benefit can be attributed to children who live long enough to receive more product and move into the high ratio and low deficit resuscitation group. In contrast to cohorts in whom data analysis includes 24 hours of resuscitation, this cohort received massive transfusion on average over 3 hours, meaning that their RBC,

plasma, and platelet transfusion happened over a condensed period in response to immediate recognition of traumatic bleeding. Finally, few children received high ratio platelet:RBC resuscitation, and therefore, the statistical analysis lacks the appropriate power to detect a mortality benefit; further multicenter studies with large cohorts should confirm or refute these findings.

CONCLUSIONS

Children with life-threatening bleeding are at substantial risk for death. High plasma:RBC ratio was associated with improved survival 6 hours post bleed. Plasma and platelet deficits that accrued during life-threatening bleeding episodes in injured children were independently associated with increased mortality at 6 and 24 hours. Multicenter clinical trials should assess whether clinicians should target ratios and deficits as optimal pediatric hemostatic resuscitation practice.

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APPENDIX 1. GROUP INFORMATION FOR THE MATIC INVESTIGATORS

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